

Audion DevOps Tools

[Русский](#) · [User Guide](#)

A portable shell for precise operations on Windows: the Linux subsystem, virtualisation, drivers and hardware, disks, networking, file associations, keys and certificates.

Why It Exists

There is a class of tasks with no good tool for them. Not "speed up Windows" and not "strip out the bloat" — but do one precise thing and know exactly what you did: install a Linux distribution from a file, switch the machine between Hyper-V and a third-party hypervisor, stop Windows Update from replacing a graphics driver, snapshot file associations before and after installing something, export a certificate together with its private key.

Each of those is done by different means: `wsl.exe` here, `bcdedit` there, a registry policy somewhere else, PowerShell commands elsewhere again. They are all documented, but scattered — and half of them require remembering what to roll back if it goes wrong.

This program gathers them in one place, with explicit parameters, risk labels, a backup before every change, and a live log.

It is not a debloater and not a collection of tweaks. Associations are not forged by defeating Windows protections, and the registry is not edited directly where a documented mechanism exists. The program wraps the standard administration tools rather than fighting the system.

Principles

Wrap what is documented. The Linux subsystem goes through `wsl.exe`. Wi-Fi profiles through `netsh wlan`. Associations through the deployment mechanism and machine-wide policy. Certificates through PowerShell over the system stores. User choice hashes and the protection around them are never bypassed by hand.

A copy before the change. The boot configuration, associations, the `hosts` file, network state, the driver store — anything that changes is snapshotted into `backup\` first.

A risk label on every command. Muted button tones separate the gentle, the ordinary, and the forceful; dangerous actions require confirmation. Paired workflows — backup and restore, export and import, block and unblock — are framed together so the way back is always in view.

The interface is deliberately technical. Buttons are short, captions compact, and the full context — what exactly will happen, what you risk, how to undo it — lives in the tooltip. Space is saved on the caption, not on the explanation.

An unsupported edition is not a target. Home editions of Windows do not guarantee that group policies apply; the program shows the edition and by default refuses to apply where it would not work.

Nothing is pulled from neighbouring folders. Every tool lives inside the project. Historical folders are read only for comparison or recovery, and only when explicitly asked.

What Is Inside

section	about
Linux subsystem	features and updates, install from the network or a file, status, backup, clone, move, delete, register disks
Virtualisation	read-only status and diagnostics, Hyper-V or third-party mode, coexistence, toggles with a boot-config backup and reboot warnings
Networking	diagnostics, backup and restore of network state, proxy, adapters, sign-in to shared folders, Wi-Fi profiles, quick modes
Hosts and Bitrix	redirecting an address to a local endpoint, endpoint detection, DNS and port status, byte-exact restoration of <code>hosts</code> from backup
Associations	snapshot and comparison of defaults, policy protection, Microsoft built-in apps — remove, restore, keep removed — reinstall blocking, change tracking
Hardware and drivers	blocking drivers from updates, graphics driver restrictions, driver store backup, HDMI and DisplayPort audio, disk inventory, recovery environment, SSD wizard
Keys	export and import of SSH keys and certificates, gathering every credential into one folder with an inventory, moving to a new machine

section	about
Maintenance	backup of command-assistant memory, documentation export, shortcuts, bundled file search

Next

- [User Guide](#) — working through the sections, and the order of operations for the dangerous subsystems.

Technical Reference

Running

```
launcher_gui.cmd
```

An ordinary launch raises the whole window as administrator — needed for deployment, the **hosts** file, network adapters, disk work, and Linux subsystem setup.

Individual entry points for one section:

```
cli\launcher_wsl.cmd
cli\launcher_bitrix.cmd
cli\launcher_default_apps.cmd
cli\launcher_association_defense.cmd
cli\launcher_hardware.cmd
cli\launcher_docs_pdf.cmd
```

They go through the same manifest and service layer as the window, so behaviour matches.

Read-only, without the elevation prompt:

```
set AUDION_GUI_NO_ELEVATE=1
launcher_gui.cmd
```

Maintenance

<code>init_folders.cmd</code>	create missing working folders
<code>cleanup_project.cmd</code>	careful cleanup
<code>cleanup_project.cmd /DRYRUN /Y</code>	show the plan, delete nothing

Cleanup keeps scripts, configuration, documentation, licences, and the folders themselves. It removes what was generated or downloaded: the runtime, the wheel store, builds, downloads, logs, reports, working-folder contents, and caches.

Verification

```
runtime\python.exe -m py_compile system_core\ui_nicegui\app.py
system_core\services\devops_tools.py system_core\core\jobs.py
runtime\python.exe system_core\ui_nicegui\app.py --smoke
runtime\python.exe system_core\doctor.py
```

Structure

<code>config\tool_manifest.yaml</code>	command tree, fields, risk metadata
<code>config\gui_settings.yaml</code>	window settings at startup
<code>config\ui_colors.yaml</code>	theme catalogue
<code>system_core\ui_nicegui\</code>	the window
<code>system_core\core\jobs.py</code>	operation runner, output decoding
<code>system_core\services\</code>	system actions
<code>system_core\cli_operation.py</code>	running an operation from the command line
<code>tools\</code>	project-local utilities
<code>backup\</code>	snapshots and rollback data
<code>logs\</code>	operation logs

The service boundary. The window holds no system logic: it collects parameters, shows risk and confirmation, and calls the operation through the manifest and service layer. System actions live in services, project modules, and project scripts; operation definitions live in `tool_manifest.yaml`.

Workbench Naming

One shared vocabulary across all Audion projects. The buttons are always named the same: **Source**, **Add file...**, **Target**, **Reset**, **Delete**, **List**. In Russian: **Источник**, **Добавить файл...**, **Назначение**, **Сбросить**, **Удалить**, **Список**.

Reset restores the project folders and deletes no files. **Delete** clears the current source and target only after confirmation. The words **Destination**, **Clear**, «Цель», and «Очистить» are not used for these controls.

Microsoft Documentation Behind It

- [Application defaults policy](#)
- [Associations through deployment servicing](#)
- [Wi-Fi profiles](#)
- [Linux subsystem commands](#)

Exports of SSH keys, certificates with private parts, and Wi-Fi passwords are sensitive: the resulting archives are stored as secrets.